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Creating a Patent System in the European Periphery: The Case of Norway, 1839-1860

Bjørn L. Basberg

Abstract

The Norwegian legislation regarding patents goes back to 1839. The first part of this article analyses the process preceding this legislation, and the development of the next twenty years when an administrative and examination system was gradually established. The creation of the patent system is viewed as an example of institution building, which at the time was considered important as a stimulus to both industrialization and modernization. The latter part of the article examines the trends in applications and grants and suggests several explanations for the very moderate patent activity of these years.

Introduction

The period from the 1840s through to the latter part of the nineteenth century is usually considered as the period of emerging industrialization in Norway.¹ The same period also saw the creation of a Norwegian patent system. The first legislation was passed in 1839, and the focus of this paper will be on the years immediately preceding and following this event.

Most economic and wider historical research on patents focuses not on the patents themselves, but on inventive activity, innovation, technological change, economic change and the process of industrialization at large. How far is it possible to use patent data to shed light on such wider issues has been debated by economists and economic historians for a number of years. Economists have been interested in the use of patent statistics as a quantitative technology indicator, and much research has been devoted to the problems and potential of such approaches.² My earlier research on the Norwegian patent records falls within this tradition, but it indicates that patent statis-

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1 For reviews of Norwegian economic history research on this period see, Sejersted, F., A Theory of Economic and Technological Development in Norway in the Nineteenth Century, *Scandinavian Economic History Review*, 1992:2, and the recent contribution by Nordvik, H. W. and E. Hovland, Det industrielle gjennombrudd i Norge, 1840-1914, med samtidens og ettertidens øyne, in Basberg, B. L., H. W. Nordvik and G. Stang (eds.), *I det lange løp. Essays tilegnet Frit: Hodne*, Bergen 1997.

2 Several review articles have shown the different perspectives of this research – from economic history, to the economics of technological change and research policy; Pavitt, K., Patent Statistics as Indicators of Innovative Activities: Possibilities and Problems, *Scientometrics*, Vol. 7, 1985:1-2, Griliches, Z., Patent Statistics as Economic Indicators: A Survey, *Journal of Economic Literature*, Vol.28, Dec. 1990; Archibugi, D., Patenting as an Indicator of Technological Innovation: A Review, *Science and Public Policy*, Vol. 19, 1992:6, and Basberg, B. L., Patents and the Measurement of Technological Change: A Survey of the Literature, *Research Policy*, Vol. 16, 1987.

tics used at an aggregate level are very difficult both to interpret and to use as a technology indicator.³ Disaggregation is necessary, and it is also necessary, in Christine MacLeod's words »to move back from measurement to judgement«.⁴ In this paper my perspective will be this latter approach, and the questions and problems raised in the Norwegian context are especially inspired by in-depth historical studies on patent system development during early industrialization in other countries.⁵ Issues related to technology-indicator questions will be focused on when consideration is given to what the Norwegian patent records may tell us about inventive activity, technological change and technology transfer in the period. The main aim though will be to describe and analyse the institutional development, the people involved and their various interests in the patent system, and so use early Norwegian patents history to shed light on early industrialization in the periphery.

There is no extensive body of literature explicitly dealing with the history of the Norwegian patent system. Most important are the histories of the institutions which managed the system at one time or another.⁶ Other historical aspects are also dealt with in works on the legal evolution of the patent system.⁷ In studies on Norwegian history more generally, including general economic histories, the development of the patent system, more often than not, is lacking.⁸ Even in the multi-volume history of Norwegian central government, the treatment of patent history is strikingly superficial. In the first volume dealing with the period to 1844 patents are not mentioned at all, while in the second volume concerned with the years 1845 to 1884, the debate and the steps leading to the further legislation of 1885 is mentioned, but as if there were no patent system already existing.⁹

This paper deals primarily with the years from 1839 to 1860, the first twenty years of Norwegian patent legislation. The longer period, from the 1820s to the 1880s will also be briefly surveyed. The origins of the legislation of 1839 can be traced back to the 1820s. The patent regime founded around 1840 was, on the other hand, to remain pretty much unchanged for forty-five years, until the patent act of 1885.

3 Basberg, B. L., *Patenter og teknologisk endring i Norge, 1840-1980. En metodediskusjon om patentdata anvendt som teknologi-indikator*, Bergen (NHH) 1984. A summary in English is found in Basberg, B. L., Patent Statistics and the Measurement of Technological Change. An Assessment of the Norwegian Patent Data, 1840-1980, in *World Patent Information*, Vol. 6, 1984:4.

4 MacLeod, C., *Inventing the Industrial Revolution. The English Patent System, 1660-1800*, Cambridge 1988, p. 5.

5 See in particular MacLeod, *Inventing*, and Dutton, H. I., *The Patent System and Inventive Activity During the Industrial Revolution, 1750-1852*, Manchester 1984.

6 Svinndal, Aa. (ed.), *Styret for det industrielle rettsvern (Patentstyret) – 50 år*, Oslo 1961. The book has an abstract in English, where especially one chapter is of interest here: Svinndal, Aa., Patent organisation in Norway. A Historical Review. See also Hasund, S., *Det Kongelige Selskap for Norges Vel, 1809-1909*, vol. II, Gjøvik 1941, Brevig, L. et.al. (eds.), *Industrielt rettsvern. Bryns Patentkontor 100 år*, Oslo 1977 and Bryns Patentkontor, *Patentvesenets samfunnsøkonomiske oppgave* (11 booklets), Oslo 1952-65.

7 Knoph, R., *Åndsretten*, Oslo 1936. See also Bryn, T., *Norsk patentlov og praksis*, Oslo 1937.

8 One exception is Hodne, F., *Norges økonomiske historie, 1815-1970*, Oslo 1981, p. 327.

9 Benum, E., *Sentraladministrasjonens historie, 1845-1884*, Vol. 2, Oslo 1979, pp. 158-62.

The first steps – and the legislation of 1839

The first Norwegian patent application was filed in 1827.¹⁰ It was not granted until 1843, a delay of some sixteen years, reflecting the lack of rules, legislation and administration. The need for a patent act was discussed in the Storting in 1824, and a committee was formed to prepare a draft based on similar legislation in other countries. It is not surprising that inspiration came from abroad. Apart from the English legislation dating from 1624 and the legislation of the 1790s in France and the United States, no other countries had any legislation on patents before 1810. From then and throughout the 1820s and 1830s a number of other countries adopted patent laws, and peripheral Norway was on the bandwagon.¹¹ The preparatory work in Norway was completed in 1826, and it became the first attempt to create a patent act in Norway.¹² It was, however, rejected by both the Storting and the government. The same fate awaited another patent act proposal in 1830.¹³

Simultaneously another work started which included the patent system as well. In 1824 a draft law was prepared about the regulation of trades in general. This was passed by the Storting in 1827. But the government was hesitant, and did not sanction the act until July 1839. It was entitled an *Act for the Protection of Trades and Crafts* (Lov om Haandværksdriften). Norway now had her first patent legislation. But it was not much of a start. The act dealt with a number of issues, but the patents were treated in only one paragraph (§ 82). It did outline, however, a basic principle which has been followed ever since: a patent was to be granted for a limited period of time. In return, the patentee had to convey a full description of the invention. The paragraph read as follows:

Patents covering discoveries and inventions in the fields of art, crafts, trades and industry may be granted by the Crown, though not for any period longer than 10 years and provided that the applicant, prior to the granting of the patent, provides an exact account of when the invention or discovery was made, and of what it consists, so that other persons will be in a position completely to exploit the discovery or invention after the expiry of the patent. If the invention has been made simultaneously by several inventors, and all want an exclusive right to use the invention, then they may all expect to receive a patent, but every one of them may as well waive the claim to the same effect as if all of them waived the claim. The one to whom a patent is granted ought, in cases where the use of the said patent is made within the boundaries of a town, to take up citizenship of that town and assume the

10 The following to a large extent is based on Svinndal, *Styret*, p. 49 ff.

11 On the early history and spread of the patent system, see for example Macklup, F., *An Economic Review of the Patent System*, Washington 1958, p. 2 ff. Precisely *where* to seek inspiration was not an obvious task. According to Machlup, as many as four different »legal philosophies« about the nature of the inventor's right existed in the first part of the nineteenth century.

12 The proposal from the committee was printed in *Stortingsforhandlinger 1833*, vol. 1, p. 522 ff. The draft was a comprehensive one, with some 30 paragraphs. With a few exceptions (see later about the so called imported patents) they expressed the principles which became the basis for later patent acts.

13 See, *Stortingsforhandlinger*, 1830, mars, p. 386. The proposal was a shorter one than the one suggested by the 1824 committee (14 paragraphs). It was also less »modern«. A patent could be obtained not only for an invention, but for a trade, i.e. an arrangement similar to a traditional privilege.

same rights and duties as master craftsmen there, but is exempt in all circumstances from the necessity of submitting any examination for the Master's Diploma.¹⁴

After the act was passed, the number of patent applications increased, and the handling of them was gradually better organized. But a single patent paragraph was obviously not sufficient to guide the people who were examining the applications. An application in 1840 from Jacob Mørch about the preservation of lobsters, especially created discussion.¹⁵ A committee was set up, and its proposals finally led to a Royal Resolution (kgl.res.) of 30 November 1841. This became an important supplement to the Act of 1839. But the government still felt uncomfortable with the legal framework. Three years later, in 1844, a new committee was set up to make proposals for a further patent act. But again there were disagreements within the government about the matter, and it was turned down. No new initiative was taken up before the 1870s. In 1876 a committee was established and its proposals resulted in the first patent law in Norway of 16 June 1885. Thus, until 1885, the act of 1839, together with the Royal Resolution of 1841, was the legal framework for the Norwegian patent system.

What about the institutional framework? Which institutions and which men examined the applications and granted the patents in these early years of Norwegian patent history? According to the act of 1839 a patent was granted by the Crown. Applications could be sent either to the county administration (Amtmann) or to the ministry, which before 1849 was the Ministry of Finance. From 1850 the responsibility for patents was exercised by the Ministry for the Interior (Indredepartementet), a department which had been established in 1845 and which became an important governmental institution in the management of the early industrialization of Norway.¹⁶ In Sejersted's words, it was a »ministry of modernisation«.¹⁷

The ministries lacked the expertise to undertake the actual examination of the applications, and a practice involving an independent group of experts was adopted during the handling of the difficult lobster preservation application of 1841. The Ministry forwarded the application to *Selskapet for Norges Vel*. This was an independent society that had been established in 1809 in an atmosphere of patriotism and nationalism. The aim was to stimulate economic development through information and financial support across a wide range of fields. The society had several expert groups, among them the so-called *Industriklassen* (Industries Group).¹⁸ This group

14 The translation follows Svinndal, *Styret*, (summary in English), p. 20, except for the sentence »If the invention (...) waived the claim«, which was deleted from Svinndal's translation. The meaning of the sentence is unclear. In Norwegian it reads: »Have flere samtidigen gjort samme Opfindelse, og samtlige ønske Eneret til Udøvelsen derav i forhold til Andre, da kunne de vente at erholde Patent paa Vilkaar som ovenanført, dog saaledes, at Enhver af dem kan frafalde denne Eneret med samme Virkning, som om Alle havde frafaldt den«.

15 See Svinndal, *Styret*, p. 54. Mørch's application concerned an improved method for the transportation of lobsters. The background was the dissatisfaction with the methods used on board British ships transporting lobsters from southern Norway to the market in London. The British companies considered Mørch's invention as a treach. They protested, and tried to convince the Norwegian authorities that his ideas were not new. The local administration in Kristiansand, however, recommended the application. The Government was put in a difficult position, since no clear rules or practice could help in the decisionmaking. The extensive correspondence on this case is in the archive of *Indredepartementet*, 2. indrekontor »A«, Udfærdigede patenter, 1842-1853 (Riksarkivet, 3A131 3/1, bx. 89).

16 Maurseth, P., *Sentraladministrasjonens historie, 1814-1844*, Vol. 1, Oslo 1979, pp. 101ff. and 414.

17 Sejersted, *A Theory*, p. 52.

18 Hasund, *Det kongelige selskab*.

was involved in the work on the lobster application, and soon became in many ways a »patent office«. The members did not just undertake an administrative handling of the applications, but did a real examination of the quality of the invention from a technological point of view. A novelty criterion was adopted, although as will be shown, there was in the beginning no general agreement about how this criterion should be defined. *Industriklassen* did not only decide whether to grant a patent or not, but the period of legal protection as well. The act said a maximum of ten years, and the practice was adopted that patents were granted for five or ten years – and later also for seven years – according to how important the invention was considered to be.¹⁹ The Ministry usually followed the recommendations of *Industriklassen*.

It is worth emphasizing that *Selskapet for Norges Vel* was a private institution. This reflected the fact not only that the central bureaucracy of the young nation was not much developed at this point, but also that co-operation between the state and the private sector was typical for Norway at this time.²⁰ It was not until the early 1870s, when the number of applications, and consequently the workload for *Industriklassen*, increased dramatically, that *Selskapet for Norges Vel* asked the government to be relieved from its work with patents.²¹ The organisation here took the initiative, *not* the government. The institutional arrangements were finally altered with the new patent law of 1885. A so-called Patent Commission was established, a government agency which represented one more step in the direction of a patent office in a modern sense.

It is interesting to note the people who were involved in the management of the early Norwegian patent system. One striking feature was the very high proportion of what we today would call society's educated elite. There were leading industrialists, politicians, economists, scientists and legal experts among the men who both planned and managed the system. This is clearly seen in the committees which prepared new legislation, and in the *Industriklassen* which handled the applications. The 1825 committee, for instance, consisted of professor Gregers Fougner Lund (economics), professor H. H. Machmann (chemistry) and mining counsellor C. A. Collett. The so-called legal committee which prepared the Royal Resolution in 1841 was appointed by *Selskapet for Norges Vel*, and its members were professor J. Keyser (physics), attorney Bruun, and Anton Martin Schweigaard. Schweigaard, who was a leading legal expert, politician and public figure, became much involved in the work with and debate about the organisation of and legislation concerning patents (see further below). Schweigaard was also involved in the preparation work leading to the patent law proposal of 1844. This work was carried out by the *Industriklassen*. But the major task for the *Industriklassen* was, as we have seen, the examination of the applications. The members were supposed to be industrial experts. They were, to borrow a phrase from Robert Post, the »scientific men«.²² In

19 In retrospect, it is, of course, possible to see that the *Industriklassen* did not always make a correct assessment about the future importance of inventions. Svinndal, *Styret*, p. 64, mentions, for example, an application from T. A. Edison in 1880 for a lightbulb. The patent obtained a seven-year protection, indicating but medium importance ...

20 Sejersted, *A Theory*, p. 52.

21 Hasund, *Det kongelige selskab*, p. 449.

22 Post, R. C., 'Liberalizers' versus 'Scientific Men' in the Antebellum Patent Office, *Technology and Culture*, Vol. 17, 1976, pp. 24ff.

the years to 1860 only six people were actively involved in this work. Only in 1856 was a full time secretary employed.²³ But the number of applications was moderate, never exceeding 20 in any year before 1860. What these people lacked in numbers, they showed in skill and status. One was a lawyer (J. H. Rye), another an army officer (F. Meilænder), while three were very well known industrialists and entrepreneurs (P. Møller, P. S. Steenstrup and J. J. Jensen). The sixth person involved in patent matters in the *Industriklassen* before 1860 was Ole Jacob Broch. Like Schweigaard he was a leading public figure at the time, a scientist (mathematician), politician and business entrepreneur.

The journal *Polyteknisk Tidsskrift* should also be mentioned. From 1854 most patents were published in this journal (when half the period of protection had expired). This was the most important journal in Norway at the time for industrial and technological developments. It is interesting to note that among the editors, were several men who occupied positions in the *Industriklassen*. On the one hand they were patent examiners, on the other hand, as editors, they published their own views, on the working of the patent system.²⁴

Why was there such a concentration of well-known people from politics, science and business? We should recognise that the contemporary Norwegian industrial and scientific milieu was very small – there were not many people who could take up the burden of all the important tasks facing the young nation on her way to becoming industrialized. The leaders of the newly emerging industries were few in number, and were involved in all aspects of industrialization. Ole Jacob Broch, who was much more than an industrialist, may serve as an illustration. He was one of the most widely involved individuals on public commissions and committees.²⁵

The controversy about »imported« patents

The patent paragraph in the act of 1839 clearly did not give sufficient guidelines to the managers of the patent system. Inevitably, there were disagreements as to how the system should be practised and how the scanty rules should be interpreted. One such disagreement, which resulted in a lengthy controversy, was the issue of so called »imported« patents (*innførselspatenter*). An »imported« patent could be granted to the person who imported or transferred an invention from abroad. While an ordinary patent could only be granted to the true inventor, this special category implied that a person other than the inventor could receive a patent. The concept – and the controversy – was known abroad. In the United States it formed part of the debate leading to the patent act of 1790. George Washington was a prime supporter,

23 The small scale of the Norwegian patent administration was not, however, that exceptional. The U. S. Patent Office employed, for example, only twelve examiners in 1861, according to Cooper, C. C., *Social Construction of Invention through Patent Management: Thomas Blanchard's Woodworking Machinery, Technology and Culture*, Vol. 32, 1991:4, p. 964.

24 See, for example, *Polyteknisk Tidsskrift*, No. 1, 1859, pp. 2, 3.

25 Seip, J. A., *Ole Jacob Broch og hans samtid*, Oslo 1971. It is a striking fact that Broch's active involvement with the patent institution is not mentioned at all in this voluminous (800 pages) biography.

but the suggestion was finally turned down. In the French patent act from the same period, »imported« patents were allowed.²⁶ This was the case too in Sweden and Denmark.²⁷

In Norway, the patent act proposal of 1826 described such a category for the first time, and used the term »invention or improvement *borrowed* from abroad«.²⁸ The discussion was continued in *Industriklassen* during the handling of Jacob Mørch's applications in 1841. In addition to his lobster-preservation application, he had submitted an application for a method of preserving food in general, which had been developed abroad. The novelty criteria was, as earlier mentioned, adopted, and consequently an ordinary patent was out of question. An alternative would be an »imported« patent. The question was brought to the legal advisory committee, and Schweigaard strongly opposed an adoption of such a category of patents. The government agreed with him, but the issue was still on the agenda when the proposal for a new patent act was prepared in 1844. Again Schweigaard argued against. This time he was joined by the other legal expert on the committee, J. H. Rye. They formed a minority, however, and the final proposal opened up the possibility of granting »imported« patents. But, as noted above, this proposal was shelved. The question of »imported« patents nevertheless formed part of the debate about the system for several years, until the final word was said by the committee of 1876. No »imported« patents were ever to be allowed or granted. So the criteria of absolute novelty was maintained over the first twenty years of Norwegian patent history, and a number of applications were turned down because the methods were known and adopted from abroad.

Why was this issue brought up at all? It is easy to see the advantage, both for the Norwegian applicant *and* for society. The applicant would get the profits (if any) from an invention made by others. For society this kind of patent would function as a channel for the transfer of technology, and thus promote domestic industrial development. In Norway, where the process of industrialisation was about to start, this was considered very important. In fact, when the Storting discussed imported patents in 1842, it was said that such patents probably could become more important than patents of genuine domestic origin. The latter category was expected to promote industrial development to a lesser extent than the transfer of patents from other countries.²⁹ On the other hand, an »imported« patent would be in conflict with the

26 Lubar, S., *The Transformation of Antebellum Patent Law*, p. 935, and MacLeod, C., *The Paradoxes of Patenting: Invention and Its Diffusion in 18th- and 19th-Century Britain, France and North America*, p. 895, both in *Technology and Culture*, Vol. 32, 1991:4.

27 The Swedish case is explored by Krantz, O., *Teknologisk förändring och ekonomisk utveckling i Sverige under 1800- och 1900-talen. Iakttagelse från patentstatistiken, Meddelande från Ekonomisk-historiska institutionen*, Lunds Universitet, Nr. 26, 1982, pp. 10 ff. For Denmark, see Nielsen, A., *Industriens historie i Danmark*, vol. III, København 1944, p. 91. In Denmark imported patents were allowed at least until about 1860. They were allowed in the monarchy, but not in Schleswig-Holstein.

28 *Stortingsforhandlinger 1833*, vol. 1, Ch. 2, §1, p. 556.

29 *Stortingsforhandlinger*, 19. august 1842, p. 383. A so-called Committee on Trades (Næringskomiteen) stated that »...det her til Lands mindre er af de egentlige Opfindelses-Patenter, der kan ventes nogen væsentlig Gavn for vor Industri, end af Forplantelsen af andre Landes Opfindelser paa norsk Grund, og at det vilde være til Gavn for at befordre deslige Foretagender, at de der ville speculere i denne Retning, ved Patent i nogen Tid beskyttedes mod Concurrence af Andre,...« . See also Knoph, *Åndsretten*, p. 201.

novelty criteria and thus the original inventor's interest. The whole controversy about this issue, both abroad and in Norway, must be understood against the background of such conflicts of interest. The patent system was in its infancy, and not only in peripheral Norway, and there was great uncertainty about which practice to adopt. As we have seen, several legal philosophies were embodied in foreign patent legislation concerning the balancing of the interests of society against those of the inventor. The choice was not obvious. A small peripheral country needed a model; so why not the French? In the Norwegian debate the French term *brevets d'importation* was used, indicating a source of inspiration.³⁰ Other countries providing inspiration were, of course, Sweden and Denmark where, as we have seen, imported patents were allowed.

Explaining the patent numbers – a first look

Table 1 indicates patenting activity in Norway from 1840 to 1860. The numbers of total applications (1), total patents (2) and patents with a domestic origin (3), were initially very low. Over the first ten years, applications were below five in most years. From 1850 there was an increase, but activity was still moderate. In 1860 the number of applications was sixteen, and the patents granted numbered eight. Six were to Norwegian citizens. Over the twenty-year period applications totalled 158, and a hundred patents were granted,³¹ most being granted to Norwegians (65 per cent). Only slowly and gradually did foreigners become involved in Norwegian patents. The first foreign patent was granted in 1848, but only after 1853 were a few patents granted annually to foreigners.

30 *Stortingsforhandlinger*, 18. august 1842, p. 382. However, when an anonymous author in 1856 (*Polyteknisk Tidsskrift*, No. 8, 30. April, pp. 113 ff.) argued against imported patents, he referred to a new French patent act where such patents had now been excluded.

31 It is obviously questionable as to what may be considered »moderate« patenting activity. Compared to Sweden and Denmark over the same period, there is no doubt that Norway was behind. In the two neighbouring countries the number of patents granted increased gradually between 1840 and 1860 from about ten to about forty per year. If the imported patents in Sweden are included, the annual number increased to about seventy. See, Krantz, *Teknologisk förändring*, p. 36, and Nielsen, *Industriens historie*, p. 96.

Table 1 Patents in Norway, 1840-1860

Year	Total Applications (1)	Total Grants (2)	Norwegian Grants (3)	Grants as a % of total-applications (4)	Grants to Norwegians as a % of total grants (5)
1840	1	0	0	0	-
1841	4	0	0	0	-
1842	5	3	3	60	100
1843	7	3	3	42,8	100
1844	3	1	1	33,3	100
1845	4	2	2	50	100
1846	2	0	0	0	-
1847	3	0	0	0	-
1848	4	3	2	75	66,6
1849	4	1	1	25	100
1850	2	2	2	100	100
1851	9	5	5	55,5	100
1852	8	7	4	87,5	57,1
1853	10	5	5	50	100
1854	10	7	3	70	42,8
1855	13	12	7	92,3	58,3
1856	17	14	10	82,3	71,4
1857	9	7	1	77,7	14,3
1858	8	7	3	87,5	42,8
1859	19	13	7	68,4	53,8
1860	16	8	6	50	75
Sum	158	100	65	63,3	65

Sources: (1) Svinndal, Aa. (ed.), *Styret for det industrielle rettsvern – 50 år*, Oslo 1961, p.66. (2) *Register over norske patenter utfærdigede indtil 1ste januar 1886*, K.ra. 1896. (3) *Register over...*, »Bekjendtgjørelse fra Departementet for det Indre« in *Polyteknisk Tidsskrift* from 1854, and the archives of Indredepartementet, 2. indrekonto »A«, *Utfærdigede patenter, 1842 – (Riksarkivet, 3A131 3/1 –)* and *Det kongelige Selskab for Norges Vel, correspondence 1849 – (Riksarkivet, 3B052 26)*.

Notes on data and sources: Several lists of applications and granted patents exist, and there are small variations between them. Svinndal used a list of applications set up by the secretary of the Patent Commission, which is considered the most accurate for this period. This is the one shown in (1). The archives of Det kongelige Selskab for Norges Vel and Indredepartementet contain applications, correspondence and the original patent documents. They do not seem to be complete, however, and the *Register over norske patenter utfærdigede indtil 1ste januar 1886* has therefore been used to compile the list of patent-grants (2). The register itself is not complete as to the addresses and places of origin of the patentees. To compile the list of granted patents to Norwegians (3), the original documents and the *Polyteknisk Tidsskrift* (where a full description of most patents was published annually) were consulted. The list of Norwegian patentees (3) varies in a few years from the list based only on the Register, which was previously published by the author (Basberg, *Patenter og teknologisk endring*).

Why did the number of patents in Norway remain so low in the first twenty years of the system's existence? As always, when interpreting patent data, one should carefully distinguish between institutional, political, economic and technological factors. There is a need also to distinguish between applications and patents granted, since, at least partially, they may be explained in different terms. There are at least four possible explanations.

First, although Norway from 1839 had a patent law, the management of the system did not function very well during the first years. In truth, there was no administration at all, and the authorities were very reluctant to deal with the applications. It was a period of trial and error. Only gradually did the management become better organized in the hands of the *Industriklassen*. In the very first years, most applications were rejected. This affected the number of grants, of course, but it might also have created a negative attitude to the system, keeping potential applicants away. There were also significant fluctuations year to year in the number of grants expressed as a percentage of applications filed (Table 1, (4)). This may indicate that the decisions of the patent authorities were difficult to predict, a fact which might also have effected the number of applications even for the later years. The percentages gradually became more stable, especially in the 1870s, and really only after 1885 when the new patent law was passed.³²

A second institutional explanation has to do with the routines for the examination of novelty. We have seen that Norway, contrary to several other countries, had adopted a novelty criterion from the start. This procedure kept down the number of patents granted compared to a system with an administrative handling of applications. The fact that imported patents were also rejected, emphasizes the strict character of the Norwegian system.³³ One example may serve as an illustration. A factory master applied for a patent in 1854. It was discovered that the actual invention was based on an American machine, and the application was rejected. The Norwegian »inventor« admitted that the reason why he had applied for a patent was that it would increase the attractiveness of the product, which he intended to manufacture.³⁴ A patent grant in this case would have become a marketing instrument more than a protection of a new invention. This, and the many other examples of rejections based on the novelty criterion, shows that the system worked.³⁵ So, on the one hand, the system did not function very well in terms of management and predictability. But on the other hand,

32 On the importance of predictability and the working of institutions which dealt with patent matters during early industrialisation, see Kahn, B. Zorina, Property Rights and Patent Litigation in Early Nineteenth-Century America, *Journal of Economic History*, Vol. 55, 1995:1.

33 Benum, *Centraladministrasjonens*, p. 158, writes that the granting of patents before the new act of 1885 was based on administrative judgement. He does not reject the fact that imported patents were not allowed, but considers the handling of other applications where, in his view, no precise criteria existed. This does not satisfactorily characterise the work done by *Industriklassen*, where actually a novelty criterion was adopted.

34 Valen-Senstad, F., *Norske landbruksredskaper, 1800-1850-årene*, Lillehammer 1964, p. 264.

35 One patent is to be found in the records which one might expect would have been rejected. In 1852 Messrs. James Small and George Hartmann of Trondheim (Norway) were granted a patent on gas-lighting for five years. The description said that the method was invented by a professor Pettenkofer in Munich, and transferred (overdragen) to the Norwegians (*Polyteknisk Tidsskrift*, 1856, p. 118). This resembles a license, not a patent.

it functioned well in terms of the adoption of the novelty criterion. Both factors had the effect of keeping the number of patents down.³⁶

A third factor which may explain the low number of Norwegian applications and grants, has to do with a lack of interest in, or direct opposition to, the patent institution for ideological and political reasons. This was the age of economic liberalism, and the period from the 1850s to the 1870s was especially marked by international debate and controversy over the system of patents which was directly linked to monopolies and privileges, and thus the values of the past.³⁷ The question is, to what extent was this debate of significance in Norway? The international controversy obviously extended to Norway, and a debate took place there as well. The leading Norwegian economist at the time was Schweigaard referred to above. He advocated liberalism, but to what extent has been debated by historians. According to Bergh and Hanisch, he had no sympathy for »Manchester liberalism« or »laissez-faire« in the extreme sense.³⁸ That might explain why he was not take on side of the critics of the patent system. On the contrary, he was, as we have seen, engaged in the attempts to improve the Norwegian patent institution from the 1840s, together with other members of the economic and political elite. It should be emphasized, however, that liberalism *had* a strong position in Norway at this time. The Act for the Protection of Trades and Crafts of 1839 was a clear example of this new trend. The act, first and foremost, dealt with reductions in regulations and the abrogation of privileges. But why, then, have paragraph on patents? Wasn't that really a striking exception from the spirit of the act?³⁹ The answer is probably that a patent was considered as *another kind* of monopoly than the traditional privileges. This was at least the case in the United States, where a debate preceded the introduction of the Patent Act of 1836. In Lubar's words, president Jackson and his followers »did their best to destroy the privileges of corporations while protecting the patent system.« The reason was that a patent was considered »a monopoly for the common man«.⁴⁰

Benum, in his work on the history of the Norwegian central government after 1845, explains the renewed interest in patent law issues in the country in the 1870s by a change to more positive attitudes towards the system in general.⁴¹ This may, of course, be part of the explanation for the developments in Norway. It is clear that the proposal from the patent committee of 1876 was an explicit defence of the patent system, which must be understood against the background of the controversy of the preceding years.⁴² But Benum does not mention any examples of former resistance in Norway concerning ideological aspects. There were examples, as have been men-

36 Any comparison with, for example Sweden which had allowed for imported patents and had not adopted the examination of novelty in the 1830s and 40s, must be undertaken with great care. The higher number of patents in Sweden in part must be explained by the fact that Norway had a more restrictive system of examination. In addition, as many as half of the patents granted to Swedish citizens were in some years imported patents. See, Krantz, *Teknologisk förändring*.

37 Machlup, F. and E. Penrose, The Patent Controversy in the Nineteenth Century, *The Journal of Economic History*, vol. 10, 1950:1 and Dutton, *The Patent System*, pp. 23-29.

38 Bergh, T. and T. J. Hanisch, *Vitenskap og politikk. Linjer i norsk sosialøkonomi gjennom 150 år*, Oslo 1984, pp. 42 ff.

39 This is suggested by Knoph, R., *Åndsarbeidets rett*, Oslo 1938, p. 38.

40 Lubar, *The Transformation*, p. 942.

41 Benum, *Sentraladministrasjonens historie*, p. 159.

42 Knoph, *Åndsretten*, p. 202.

tioned, but we will argue that the leaders of opinion, with Schweigaard in the van, were in this matter never part of the strong opposition. The developments of the 1870s were really a continuation of the efforts to establish workable patent legislation, which had been going on for thirty years. The low number of patents before 1870 has to be explained in other terms. If opposition was a factor, it was probably chiefly of another kind, and the product of the inefficient way the patent system was managed in these early years. This was, in fact, not only true for Norway during the patent controversy. It has also been observed by Dutton in his study of the British experience in these years.⁴³

A fourth possible explanation is that the low number of patents might have reflected poor Norwegian inventive activity. In the introduction above was raised the well-known questions about this relationship. There are, in general, a number of problems of interpretation, and they definitely seem to be present in these early years of Norwegian patent history. The number of Norwegian patents was very low, and several reasons have been mentioned for this, which do not necessarily have to do with little inventive effort or interest. However, in analysing the patent-invention relationship one important issue is the *quality* of the patents. As we have seen, a most important precondition for a reasonably high quality was at work; the novelty criterion was practised in the examination of applications. When dealing with the patents of Norwegian origin, it is also important to remember that »imported patents« were *not* granted, thus preventing a category of patents which had no connection with genuine domestic inventive activity. We have also argued that the so-called patent controversy did not significantly affect the attitudes of Norwegian inventors. So it is our view that, both the low number of applications and the number of grants must, to some extent, reflect a moderate inventive activity. Norway was about to enter the age of industrialization, but was clearly in an infant – if not an embryonic – stage. The industrial sector as a whole still made only a moderate contribution to the economy.⁴⁴

Patents, industrialization and the transfer of technology

If patents reflect inventive activity, it is of interest to investigate more closely the areas of interest to inventors. Were the patents concentrated in any specific areas, and what was the relationship to the emerging industrial areas and the areas of technology transfer in the period? Instead of investigating every single patent, it is possible to investigate categories of patents which might give a better indication of trends or technological clusters of interest. This could be done in several ways. The patents granted before 1886 were categorized in as many as 531 groups.⁴⁵ Most categories contained only one patent, and it is difficult to use them to get an overview of the most important areas of patenting activity. The well-known patent classification system of 89 classes was not introduced until 1885, and is consequently of no use for the period we are investigat-

43 Dutton. *The Patent System*, p. 26.

44 Hodne. *Norges økonomiske historie*, p. 46, Bergh, T. et.al., *Norge fra u-land til i-land. Vekst og utviklingslinjer 1830-1980*, Oslo 1983, p. 144.

45 See *Register over norske patenter utfærdigede indtil 1ste januar 1886*, Kr.a. 1896.

ing.⁴⁶ A third possibility is to distribute the patents by industry. This is not an obvious procedure, and I have in earlier works put some effort into a procedure of categorizing the Norwegian nineteenth and twentieth century patents by industrial sectors.⁴⁷ However, when dealing with the very early years of Norwegian patent history, when the numbers were low, the industrial sectors are so wide that too much information is hidden by such categories. The following therefore takes a middle-of-the-road approach. A number of patents can be easily categorized by an industry or industrial sector. Those that cannot, are not categorized in broad sectors, but are listed according to more narrow technological areas. Table 2 shows the Norwegian patents from 1840 to 1860 categorized along such lines. The dominating patent sectors were the sectors which were also important to the Norwegian economy at the time; the sawmill industry (wood-processing), agriculture, fisheries and shipping.

Table 2 Patents in Norway by industry or technological areas, 1840-1860

Sector	Grants Total	Grants Norwegians
Handling of wood (sawing, planing, wood-working, wood-oil)	12	11
Fisheries (equipment and processing)	9	7
Agriculture (equipment, fertilizers)	7	6
Ship design and construction, navigation	7	5
Water-power and -supply (turbines, water-wheels)	7	6
Gas production and gas lighting	6	4
Metallurgy, mining	6	3
Electricity, telegraph	6	0
Firearms, ammunition, explosives	6	2
Ovens, drying	5	2
Tobacco machines	4	4
Transportation	3	3
Match making	3	2
Drills, drilling machines	3	3
Household equipment	3	2
Mills	2	1
Textiles, apparel	2	0
Brick and tile making	2	0
Chemical products (sulphur)	1	0
Paper pulp	1	0
Other machinery	5	4
	100	65

Source: *Author's classification.*

⁴⁶ Svinndal, *Stryet*, has categorized the patents in 89 classes for the period before 1885, but only as a total for each class in the period 1840-85.

⁴⁷ Basberg, *Patenter og teknologisk endring*.

Among the patents categorized by technological area, waterpower and watersupply stand out. It is also interesting to note the country of origin of the patentees in the different sectors. In the largest sectors, which were also the important sectors in economic terms, the majority of patentees were Norwegians. In the sectors which reflect newly emerging economic and technological areas, the patents were fewer, and were dominated by foreign patentees (chemicals, electricity, telegraph, paper pulp).

There are, of course, many possible pitfalls, when trying to interpret such sectoral variations in patenting. There has never been consistent patenting habits or policy across industries and sectors, and the variations were probably no smaller in the early years. If in one industry the patent system was considered important, in another it was not. But both industries could be characterized by high inventive and innovative activity. An industry where we might expect to find patents was the emerging textile industry. Together with the mechanical engineering industries (with patents spread over several sectors), it progressed much over the period, and its technology was to a large extent imported from other European industrial centres.⁴⁸ However, the textile industry was not among the patent-intensive sectors in this period. In fact, only two patents were granted, to the American Otis Avery for a sewing machine (1855) and the Italian Gaetan Bonelli for a loom (1854). In the following years, 1857 to 1870, Bruland found only five Norwegian patents that were related to textiles. From this and other evidence she concluded that »domestic innovative activity in textiles appears to have been low«.⁴⁹ So why was not a substantial role for imported technology reflected in the Norwegian records? One possible explanation is that the imported technology was already old and established, and was thus not possible to patent. Another explanation has to do with the patent experience of this industry. We know from research on the early British textile industry that a considerable number of inventions was never patented.⁵⁰ This practice would probably also apply in relation to the transfer of such technology abroad. Consequently, the moderate Norwegian patenting activity seems only to a very limited extent to reflect what was going on in terms of technology transfer.

This lack of a relationship between patenting and the transfer of technology may be analysed in more detail. Table 1 (5) shows the proportion of the Norwegian patents which were granted to Norwegian citizens between 1840 and 1860. In the first ten years or so, patents with a Norwegian origin dominated. The first foreigner to receive a Norwegian patent was Francois Meldon of Sussex, England, for an invention in metallurgy (1848). Three more years then passed without any grants to foreigners. But the 1850s were different. The number of patents was still low and the annual variations were substantial, but relatively more patents were granted to foreigners. For the twenty years from 1840 to 1860 35 per cent of Norwegian patents were granted to foreigners. From 1853 to 1860 (when the patents to foreigners were

47 Basberg, *Patenter og teknologisk endring*.

48 Bruland, K., *British Technology and European Industrialization. The Norwegian Textile Industry in the Mid Nineteenth Century*, Cambridge 1989, and Bruland, K., *The Norwegian Mechanical Engineering Industry and the Transfer of Technology, 1800-1900*, in Bruland, K., (ed.), *Technology Transfer and Scandinavian Industrialisation*, Oxford 1991.

49 Bruland, *British Technology*, p. 60.

50 Griffiths, T., et.al., *Inventive Activity in the British Textile Industry, 1700-1800*, *Journal of Economic History*, Vol. 52, Dec. 1992, pp. 884-85.

granted annually) the proportion was 45 per cent. Such proportions were lower than in the years to come, when the foreign share of total Norwegian patents typically was in the 60-80 per cent range. Compared to such later developments, patents from abroad did not play a very prominent role in the first twenty years of Norwegian patent history. Thus, the assumed importance of technology import in these years is not clearly reflected in the patent records.

The number – and proportion – of patents of foreign origin was low, but what about their quality? It is often assumed that foreign patents were of a higher quality than patents of domestic origin. If this was the case in the period under investigation, it may to some extent alter the impression of but a moderate influence for foreign technology. The quality-issue is not easily testable. But one approach could be to analyse how the patents were distributed across the different categories of maximum duration of protection. As we have seen, these categories were used as ways of distinguishing between differences in quality. With the exception of the engineer, N.H. Bruun, from Christiania, who received a *three* year protection in 1855 for his patent on a drilling machine, all the other patents were granted for either five or ten years. Table 3 shows how the patents were distributed between these two categories. The majority (54 per cent) was granted for five years. Among these patents, 16 (30 per cent) were granted to foreigners. Foreigners were responsible for 41 per cent of the ten-year patents. As we have shown earlier, the foreigners' share of the total number of patents from 1840 to 1860 was 35 per cent. On the other hand; while Norwegian residents had 65 per cent of all the patents, they had 69.8 per cent of the five year and 59 per cent of the ten year patents. Consequently, if we accept the duration of the patent protection as some indication of quality, the patents with a foreign origin were on average of a higher quality than the patents granted to Norwegians.⁵¹

Table 3 Length of protection for patents registered in Norway, 1840-1860

Resident country of patentee:	Five-year protection	Ten year protection
Norway	37 (69,8%)	26 (59%)
Other countries	16 (30,2%)	18 (41%)

Source: See Table 1, (2) and (3).

Note: The patents in this table add up to 97, while Table 1 lists 100 patents. Three patents are not included in this table: One patent was protected for three years, while for two it has not been possible to find the duration of protection.

On the other hand, it is not possible from these calculations to conclude that the domestic patentees were dominated by amateur inventors. The opposite actually seems to be the case. The curious and odd inventions of which we generally find

⁵¹ A referee remarked that it seemed to be easier for a Danish resident to receive a patent in Denmark of maximum duration compared to a foreigner. I have seen no such indications for Norway. If, on the other hand, it was the case in Norway, the differences between patents of foreign and domestic origin would be even more visible. A typical characteristic of most patent applications, from Norwegians as well as foreigners, was that the inventors applied for a ten-year protection. In more than half of all cases, this was reduced by the patent authorities.

numerous examples in patent records, are very largely absent in the years under consideration in this study. For example, only two patents were granted to Norwegians for household equipment (knives) – a typical invention of the amateur inventor. The typical Norwegian patentees were craftsmen and engineers, and we also find several leading industrialists among the most active users of the patent system.⁵²

Conclusion

This paper has studied the creation of the Norwegian patent system. It was but a small institution in a small country. It has been argued, however, that it is worth studying in the context of the early Norwegian industrialization process, and that it deserves rather more attention by economic historians than so far has been the case. The creation of a patent system provides an example of institution-building which has importance for development and modernization. The Norwegian controversy over the so-called »imported« patents, for example, reveals an eagerness in creating mechanisms for technology transfer – a paramount issue for the young nation. The way the system was organized and managed, involving both the state and private institutions, illustrates the Norwegian way to modernization. The development and running of the system involved many people belonging to the industrial, political and scientific elite. So the system could not have been entirely in the periphery of the periphery. The patent system was, however, obviously of marginal importance only in the first twenty years of its existence. It took time to make the new legislation of 1839 work, and it took time to find an efficient way of managing the system. The patent institution received a moderate interest, from both foreigners and Norwegians. The »imported« patents were never accepted. The number of applications stayed low until the 1870s, when they increased sharply. There was then a »take-off« in patenting, more than a hundred years later than had been the case in Britain.⁵³

52 A number of issues concerning single patents and the men behind them may be worth pursuing. One such issue, regarding a broadening of the segment of the population who took part in patenting, is discussed by Sokoloff, K. L. and B. Zorina Khan, *The Democratization of Invention During Early Industrialization: Evidence from the United States, 1790-1846*, *The Journal of Economic History*, Vol. 50, 1990:2. The concept of the »quasi-professional inventor« raised by Dutton, *The Patent System*, p. 108 ff., is another issue which could also be analyzed in the Norwegian context as well.

53 MacLeod, *Inventing*, pp. 144ff. and R. J. Sullivan, *England's »Age of Invention«: The Acceleration of Patents and Patentable Invention during the Industrial Revolution*, *Explorations in Economic History*, Vol. 26, 1989. Sullivan estimates a distinct acceleration in 1757. The increase in Norway was first of all caused by an increase in the number of applications from abroad. The patent controversy came to an end, and the 1870s was a period of a more positive attitude to the system internationally, and consequently the number of Norwegian applications, especially from abroad, increased. The increase in Norwegian domestic applications may be related to the increased technological vitality in the country in this period, a trend associated with a shift from a demand- to a supply-side stimulated development (see Sejersted, op.cit.). Whether or not the patent statistics may be used to shed light on such questions is not the topic of this paper, but is the centre of focus of the well-known Schmookler-type approach to patent research, which still inspires researchers. See, for example Sokoloff, K. L., *Inventive Activity in Early Industrial America: Evidence From Patent Records, 1790-1846*, *Journal of Economic History*, Vol. 48, 1988:4, pp. 813 ff.

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Questions concerning the interpretation of patent data are complicated, and I have only touched on some of the ways we can approach them. It is obviously correct that patents »are 'fuzzy objects', like the peaks in a mountain range«. ⁵⁴ I would, however, like to conclude by citing another metaphor from the same field: »Yet, like the mountain, patent data are there, and cannot be ignored«. ⁵⁵

⁵⁴ MacLeod, *Inventing*, p. 7.

⁵⁵ Lawson, W. S., The Use of Patent Information as an Indicator of Technological Output, OECD Workshop on the Measurement of R&D Output, Paris 1979, p. 10.